

SoilBR-Onto: An Ontology for Soil Fertility Management and Classification in Brazil

Helfer, Gilson Augusto; da Costa, Adilson Ben; Barbosa, Jorge Luis Victória (2025.0)

ABSTRACT ORIGINAL

Soil analysis laboratories require extensive data for precision agriculture, including parameters such as texture, organic matter, phosphorus, and potassium, which guide fertilization and soil acidity correction through pH. This article presents SoilBR-Onto, an ontology that advances the use of fertilizers, liming recommendations and classification of Brazilian soils based on laboratory analysis. A total of 481 axioms and 145 semantic rules enable accurate queries and inferences using real soil sample results. The SoilViewer Mobile application, developed within this study, facilitates access to fertilization and liming recommendations and enables the visualization of soil deficiencies through heat maps. Furthermore, this ontology improves soil data management and decision-making, applicable beyond applied tobacco, to crops such as soybeans, wheat and corn, promoting environmental sustainability and efficiency in agriculture. © The Author(s) 2025